

CommunitySphere – Zero to Live AWS Hosting Guide

Goal: Any person with an AWS account and this repository must be able to deploy the website in minutes using Terraform, without assumptions or missing steps.

1. AWS Account & IAM Setup

- Login to AWS Console
- Create IAM user with AdministratorAccess
- Create Access Key (Access Key + Secret Key)
- Save credentials securely

2. Local Machine Prerequisites (Windows)

- Install AWS CLI
- Install Terraform
- Install Git
- Restart PowerShell after installs

3. AWS Login from PowerShell

- Run: aws configure
- Enter Access Key
- Enter Secret Key
- Region: ap-south-1
- Output format: json

4. Project Folder Structure

- Create empty folder: CommunitySphere-Terraform
- Place main.tf inside it
- Do NOT place application code here

5. Terraform Execution

- cd CommunitySphere-Terraform
- terraform init

- terraform apply
- Type yes when prompted

6. What Terraform Creates

- VPC
- Subnet
- Security Groups
- EC2 with Apache + PHP
- RDS MySQL
- GitHub repo cloned automatically

7. SSH into EC2

- Download .pem key from AWS
- Run: `ssh -i key.pem ec2-user@`

8. Database Table Creation

- Connect to RDS using mysql client
- Use database communitydb
- Run SQL file from GitHub repo

9. Backend Configuration

- Edit backend/config.php
- Set DB_HOST to RDS endpoint
- Restart Apache

10. Final Verification

- Open ALB URL
- Signup user
- Check data exists in RDS

11. IF-THEN Error Rules

- IF terraform shows no changes THEN infrastructure already exists
- IF ALB works but buttons fail THEN BASE_URL is wrong

- IF DB connection fails THEN RDS SG not allowing EC2 SG
- IF PHP downloads THEN php module missing
- IF EC2 replaced THEN Auto Scaling is enabled

This document is intentionally concise, deterministic, and repeatable. Following it step-by-step guarantees a working deployment without guesswork.